Report

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1 Lab 3: Clustering, Association Rules, and Recommenders

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1.1 1. Datasets, Data Cards, and Ethics

1.1.1 1.1 Clustering Dataset: Health & Lifestyle (Synthetic)

Source / URL: Kaggle – "Health & Lifestyle Dataset"
License: CC0 (Public Domain)
Shape: 100,000 rows × 16 columns

Variables (subset):

- id: Integer participant identifier
- age: Age in years (approx. 18–90+)
- gender: {Male, Female}
- bmi: Body Mass Index (kg/m²)
- daily_steps: Number of steps per day
- sleep_hours: Hours of sleep per night
- water_intake_l: Daily water intake (liters)
- calories_consumed: Estimated daily calories (kcal)
- smoker, alcohol, family_history: Binary lifestyle / history indicators (0 = No, 1 = Yes)
- resting_hr, systolic_bp, diastolic_bp: Vital signs
- cholesterol: Total cholesterol (mg/dL)
- disease_risk: Binary label {0 = Low risk, 1 = High risk} (used only for external evaluation)

Ethics and Provenance Note

This dataset is synthetically generated, so no real individuals are represented. That removes privacy concerns but introduces other limitations:

- Joint distributions and risk patterns reflect the generator's assumptions rather than real clinical data.
- Relationships (e.g., between BMI and disease_risk) may be exaggerated or simplified.
- The dataset is appropriate for teaching and benchmarking algorithms, but not for clinical inference or policy decisions.

1.1.2 1.2 Transactional Dataset: Online Retail

Source / URL: UCI Machine Learning Repository – "Online Retail"
License: CC BY 4.0
Shape: 541,909 rows × 8 variables

Basic counts (after loading):

- Unique customers (non-null CustomerID): 4,372
- Unique items (StockCode): 4,070
- Total interactions (rows): 541,909

Key variables:

- InvoiceNo: Transaction ID; codes starting with "C" are cancellations
- StockCode: Product code (item ID)
- Description: Product name
- Quantity: Units purchased per row
- InvoiceDate: Timestamp
- UnitPrice: Price per unit (£)
- CustomerID: Unique customer identifier
- Country: Customer's country

Ethics and Provenance Note

This is real transaction data from a UK-based online retailer (2010–2011):

- Customers include both individual shoppers and wholesalers, so behavior is mixed.

- No demographic attributes (age, income) are included, so any inferred segments may conflate unobserved factors.

- Many transactions have missing CustomerID, limiting user-level behavior analysis.

- Results are specific to this retailer, time period, and product mix, and should not be overgeneralized.

1.2 2. Exploratory Data Analysis

1.2.1 2.1 Clustering Dataset EDA

Univariate patterns

Histograms of numeric features (age, bmi, daily_steps, sleep_hours, water_intake_l, calories_consumed, vitals, cholesterol) show:

- Skewed distributions for lifestyle variables such as daily_steps, calories_consumed, and water_intake_l, suggesting heterogeneous lifestyle behaviors.

- Physiological variables (blood pressure, resting_hr, cholesterol) fall in reasonable adult ranges but do not form distinct modes.

- Categorical variables (gender, smoker, alcohol, family_history) are moderately imbalanced but both classes are well represented.

Bivariate relationships

- Pairplots of bmi, daily_steps, sleep_hours, and water_intake_l, colored by disease_risk, show only weak visual separation: high-risk and low-risk points largely overlap in feature space.

- A boxplot of BMI vs smoker indicates slightly higher BMI among smokers, but not a sharp distinction.

Correlation structure

A correlation heatmap of numeric features reveals:

- All pairwise correlations are extremely close to zero (roughly between –0.01 and +0.01), including between cholesterol and systolic_bp.

- The data behave almost as if features are independent, which is consistent with a synthetic generator that samples variables separately or with weak coupling.

- There is no meaningful multicollinearity; therefore, no numeric variables need to be removed or combined purely to address correlation.

Scaling implications

Although correlations are negligible, the numeric features are on very different scales:

- daily_steps can be on the order of 10,000–20,000,
- blood pressure and heart rate variables are around 60–180,
- bmi is typically 15–45, etc.

To prevent large-range features (like daily_steps) from dominating Euclidean distances, all numeric features are standardized before clustering.

1.2.2 2.2 Transactional Dataset EDA

After dropping rows with missing CustomerID and StockCode and renaming fields to user_id, item_id, and invoice_id:

Item popularity

- The distribution of item counts (per item_id) is highly skewed.
- A small subset of products accounts for a large fraction of transactions.
- A bar chart of the top 20 items by interaction count shows a long head and a very long tail, which is typical of retail data and ideal for association rule mining.

User activity

- The histogram of interactions per user (user_id) is long-tailed.
- Most users make only a few purchases, while a minority of "heavy" customers generate many transactions.
- This suggests natural segmentation by engagement level (e.g., occasional vs frequent buyers).

Item co-occurrence

- By constructing baskets at the invoice_id level and examining co-occurrence among the top 5 items, several item pairs appear together frequently within the same transaction.
- This indicates product affinities and is a strong motivation for both association rules and recommender systems.

EDA Takeaways

- The clustering dataset looks like a mostly continuous cloud of health and lifestyle variables with only subtle hints of segmentation.
- The transactional dataset clearly exhibits heavy-tailed popularity and co-occurrence patterns that are promising for association mining and recommendations.

1.3 3. Preprocessing and Feature Engineering for Clustering

Feature selection and missingness

- Features used: four categorical (gender, smoker, alcohol, family_history) and ten numeric (age, bmi, daily_steps, sleep_hours, water_intake_l, calories_consumed, resting_hr, systolic_bp, diastolic_bp, cholesterol).

- Rows with missing values in any of these features were dropped. The dataset is large (100k rows), so dropping a small number of rows is acceptable and avoids assumptions required for imputation.

Encoding and scaling

- Categorical variables are encoded using OneHotEncoder with drop="first" to avoid redundant dummy variables.

- Numeric features are standardized using StandardScaler to zero mean and unit variance.

- A single ColumnTransformer combines these steps into a reproducible preprocessing pipeline.

- The resulting design matrix X_processed has shape (n_samples, n_features_encoded) and is used as input for all clustering methods.

PCA for visualization

- A 2D PCA projection on X_processed shows that the first two principal components explain about 18–19% of the variance.

- This relatively low variance share is expected, because features are nearly uncorrelated and each contributes modestly.

- The PCA scatter plot shows a diffuse, overlapping cloud of points without clear axes of separation, reinforcing the idea of weak cluster structure.

- PCA is thus primarily used for visualization and plotting cluster assignments in 2D, rather than for dimensionality reduction.

Distance / similarity matrices

- Euclidean distance (on standardized features) was used for K-means, HAC, and DBSCAN.

- A Euclidean distance matrix of a 1,000-point subsample confirms a well-defined pairwise structure and is computationally manageable.

- Cosine similarity was also computed on the same subsample as an alternative affinity measure; it is more typical for high-dimensional text-like data but is used here to support potential spectral clustering.

1.4 4. Clustering Methods and Evaluation

All clustering models were trained on the standardized, one-hot encoded feature matrix X_processed. The disease_risk label was never used as input, only for external evaluation with ARI and NMI.

1.4.1 4.1 K-means

- Cluster counts evaluated: k = 2, 3, …, 8.
- Internal metrics (on a 5,000-sample subset):
  - Silhouette score (higher is better)
  - Davies–Bouldin index (lower is better)
  - Calinski–Harabasz index (higher is better)
- External metrics (using disease_risk on the same subset):
  - Adjusted Rand Index (ARI)
  - Normalized Mutual Information (NMI)

Findings

- Silhouette and Calinski–Harabasz are highest at k = 2 and degrade as k increases.
- Davies–Bouldin improves slightly with more clusters but not dramatically.
- ARI and NMI remain very close to zero for all k, indicating that K-means clusters have almost no alignment with the disease_risk labels.
- Overall, K-means suggests that if any segmentation exists, it is a very weak 2-cluster structure.

1.4.2 4.2 Hierarchical Agglomerative Clustering (HAC)

- Linkage methods: Ward and average.
- Cluster counts evaluated: k = 2–8 on the X_metrics subsample.
- Metrics: same internal and external metrics as K-means.

Findings

- For both Ward and average linkage, the silhouette score peaks at k = 2 and drifts toward zero or negative values as k increases.
- Dendrograms (on a 300-point sample) show no obvious, large jumps in linkage distance that would indicate natural group splits.
- ARI and NMI remain near zero, again showing no real correspondence to disease_risk.

- HAC essentially agrees with K-means that the data are weakly clusterable and best represented by a coarse, 2-cluster partition.

1.4.3 4.3 Gaussian Mixture Models (GMM)

- Components evaluated: k = 2–8, covariance_type = "full".
- Metrics: silhouette, Davies–Bouldin, Calinski–Harabasz, ARI, NMI, plus AIC and BIC.

Findings

- Silhouette remains small and best at k = 2, confirming overlapping clusters.
- Davies–Bouldin and Calinski–Harabasz do not show strong preference for higher k.
- BIC decreases initially and may favor more components (e.g., k around 6–7), but this reflects density approximation rather than well-separated clusters.
- ARI and NMI stay near zero, so even multiple Gaussian components do not recover disease_risk.

1.4.4 4.4 Density-Based, Spectral, and Graph Clustering

DBSCAN

- Explored eps in {0.5, 1.0, 1.5, 2.0, 3.0, 4.0} and min_samples in {3, 5, 10} on a 2,000-point subset.
- Most parameter settings either labeled all points as noise or merged everything into one cluster.
- The best silhouette (~0.56) occurred at eps = 1.5, min_samples = 3, but this model found only 5 tiny core clusters and labeled 1,985 out of 2,000 points as noise.
- This high silhouette is thus an artifact of a few isolated points, not a meaningful global segmentation.

Spectral clustering

- Used an affinity="nearest_neighbors" kNN graph with 10 neighbors.
- Evaluated k = 2–8 clusters with internal and external metrics.
- The best silhouette occurs at k = 2, but values remain low and external metrics are near zero.
- PCA plots show overlapping clouds similar to K-means and GMM.

Graph clustering (connected components)

- Built a 10-nearest-neighbor graph and considered connected components as graph clusters.

- The graph was fully connected and produced exactly one component.
- Internal metrics are undefined in this degenerate case, and the PCA projection shows a single-colored cloud.

1.4.5 4.5 Overall Clustering Summary

Across all methods (K-means, HAC, GMM, DBSCAN, Spectral, graph-based):

- The data behave like a single, roughly isotropic cloud with no strong, naturally separated clusters.
- Internal metrics consistently favor k = 2 as the "least bad" partition, but even this is weak.
- External metrics (ARI, NMI) indicate almost no alignment between cluster assignments and disease_risk.
- In this synthetic health dataset, disease risk appears more continuous than cluster-based, and unsupervised clustering is not an effective way to reconstruct the label.

1.5 5. Association Rules: Frequent Itemsets and Rule Evaluation

All association analysis uses the Online Retail dataset after:

- Dropping rows with missing CustomerID
- Removing returns (Quantity < 0)
- Using only InvoiceNo and StockCode to build baskets

Each invoice is treated as a transaction, and each StockCode as a binary item in a basket.

1.5.1 5.1 Frequent Itemsets: Apriori vs FP-Growth

Basket construction

- A transaction–item matrix (basket) was built where rows are invoices and columns are StockCodes.
- Values were converted to Booleans (True if the item appeared in the transaction) as required by mlxtend.

Item filtering

- Per-item support was computed as the fraction of baskets containing each item.
- Only items with support 2% were retained, reducing the item dimension from 3,665 to 209.
- The reduced basket has 18,536 transactions $\times$ 209 frequent items.

Apriori

- Minimum support thresholds: 0.02 and 0.03.
- Maximum itemset length: 3.
- At 2% support: 251 frequent itemsets, runtime ~0.06s, max length = 3.
- At 3% support: 94 frequent itemsets, runtime ~0.02s, max length = 1 (only singletons).

FP-Growth

- Same supports and max_len as Apriori.
- At 2% support: 251 itemsets, runtime ~1.13s, max length = 3.
- At 3% support: 94 itemsets, runtime ~0.22s, max length = 1.

Comparison

- For both support levels, Apriori and FP-Growth return identical numbers of itemsets and the same maximum itemset lengths.
- Increasing support from 2% to 3% sharply reduces the number of itemsets and removes all multi-item sets.
- On this moderately sized, filtered dataset, Apriori is actually faster than FP-Growth, likely because:
  - the item space has already been pruned (support   2%), and
  - max_len is capped at 3.

FP-Growth does not offer an advantage under these conditions; both algorithms produce the same patterns.

1.5.2 5.2 Rule Generation and Evaluation

Rule generation

- Rules were generated from the 2% support itemsets (the only set containing multi-item itemsets).
- association_rules was applied with metric="confidence" and min_threshold = 0.3.
- This produced 79 rules with an average total rule length (antecedent + consequent) of about 2.08 items.
- At 3% support, no multi-item itemsets exist, so no rules can be generated.

Top rules by different metrics

- Rules were ranked by lift, confidence, leverage, and conviction.
- The top-10-by-lift plot shows that many of the strongest rules involve three item codes: 22697, 22698, and 22699.

Meaningful, business-relevant rules

Several high-lift, high-confidence rules involve the same product trio:

- {22698} → {22697, 22699}:
  Customers who purchase 22698 frequently also purchase the other two items. A lift of roughly 24 suggests a very strong association, consistent with a 3-piece set.
- {22697, 22699} → {22698}:
  The reverse combination shows that when customers buy the other two items, they almost always include 22698, forming a closed product family.
- {22699} → {22697} and {22697} → {22698}:
  These pairwise rules indicate strong "frequently bought together" relationships that are ideal for bundled promotions or "customers who bought this also bought…" recommendations.
- {22699, 22698} → {22697}:
  This rule "completes the set," reinforcing that these three items function as a cohesive family that should be stocked and marketed together.

Taken together, these rules highlight a highly actionable product bundle suitable for cross-selling, kitting, or targeted offers.

Misleading or spurious rules

Two classes of rules require caution:

- Rules with high confidence but lift close to 1:
  These rules often reflect the standalone popularity of consequents rather than genuine association. They may look strong numerically but add little new information for cross-selling.
- Rules with very low support:
  Rare-item rules can show high lift and confidence due to small denominators but are unstable and not representative of general shopping behavior. They are not attractive for marketing actions.

Threshold sensitivity

- At 2% support and 30% confidence, the rule set is rich (79 rules, many multi-item) and average rule length is about 2.1 items.
- At 3% support, all multi-item itemsets disappear, and the number of rules drops to zero.
- Raising confidence would further reduce the number of rules and bias the remaining patterns toward very popular items, improving interpretability but losing exploratory value.

Overall, a support threshold of 2% provides the best trade-off between richness of patterns and interpretability for this dataset.

1.6 6. Conclusions: Key Takeaways, Risks, and Next Steps

1.6.1 Key Takeaways

- The synthetic health and lifestyle dataset does not exhibit strong natural clusters.
  - All clustering methods (K-means, HAC, GMM, DBSCAN, spectral, and graph-based) agree on a very weak 2-cluster structure.

  - Internal metrics are modest at best, and external metrics (ARI, NMI) indicate that clusters do not meaningfully align with the disease_risk label.

  - Disease risk in this dataset appears to vary smoothly rather than forming distinct segments.
- In contrast, the Online Retail dataset contains clear and actionable association patterns.
  - Strong co-purchasing signals emerge, particularly for a specific trio of products (StockCodes 22697, 22698, 22699).

  - These associations are highly actionable for bundling, recommendation systems, and merchandising.
- Threshold tuning is crucial for association mining.
  - A 2% support threshold captures meaningful multi-item patterns, while 3% support collapses all itemsets to singletons and eliminates rule generation.

  - Confidence thresholds control the balance between novelty and interpretability.

1.6.2 Risks and Limitations

- The clustering analysis is constrained by the synthetic nature of the health dataset and by the near-independence of its features. Real-world clinical data might show more structure.

- Clustering results are sensitive to scaling, choice of distance metric, and high dimensionality; alternative feature engineering could change conclusions.

- Association rules are susceptible to popularity-driven artifacts (lift 1.0) and to unstable low-support rules. Both require careful filtering and domain judgement.

- In this particular application, FP-Growth did not outperform Apriori due to prior filtering and a cap on itemset length; conclusions about algorithm efficiency may not generalize to very large or sparser datasets.

1.6.3 Next Steps

- For clustering and health modeling:
  - Explore nonlinear embeddings (UMAP, t-SNE) to check for subtler manifold structure.

  - Engineer composite features (e.g., risk scores, ratios) that may reveal more meaningful clusters.

  - Shift from unsupervised clustering to supervised models for predicting disease_risk directly.

- For association rules and retail analytics:
  - Incorporate price, quantity, and time (seasonality, recency) into the analysis.
  - Build a simple recommender system that uses discovered rules to generate product suggestions and, if possible, evaluate them via held-out interactions.
  - Combine association rules with customer-level segmentation (e.g., heavy vs light buyers) to design personalized promotions.

Overall, this lab shows that unsupervised methods are highly sensitive to the underlying data structure: clustering was only marginally informative for the synthetic health dataset, while association rule mining yielded clear, business-relevant insights on the retail dataset.